

## **DAA – 16 Mark Exam Answers (Remaining Important Questions)**

### **1. Asymptotic Notations and Properties**

Introduction:

Asymptotic Notations and Properties is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

### **2. Recursive and Non-Recursive Algorithms**

Introduction:

Recursive and Non-Recursive Algorithms is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

### 3. Algorithm Efficiency

Introduction:

Algorithm Efficiency is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

### 4. Exhaustive Search

Introduction:

Exhaustive Search is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 5. Multiplication of Large Integers

Introduction:

Multiplication of Large Integers is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 6. Optimal Binary Search Tree

Introduction:

Optimal Binary Search Tree is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 7. Multistage Graph

Introduction:

Multistage Graph is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 8. Coin Changing Problem

Introduction:

Coin Changing Problem is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## **9. Warshall and Floyd Algorithms**

Introduction:

Warshall and Floyd Algorithms is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## **10. Maximum Flow (Ford-Fulkerson)**

Introduction:

Maximum Flow (Ford-Fulkerson) is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 11. Bipartite Graph

Introduction:

Bipartite Graph is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

## 12. N-Queen Problem

Introduction:

N-Queen Problem is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.

### **13. P, NP, NP-Complete and NP-Hard Problems**

Introduction:

P, NP, NP-Complete and NP-Hard Problems is an important topic in Design and Analysis of Algorithms. This answer includes definition, explanation, algorithm, example, complexity, advantages, applications and conclusion.

Definition:

Provide the formal definition and explain the purpose of the technique/problem.

Algorithm:

Write the step-by-step algorithm in exam format and explain each step.

Example:

Include a suitable worked example with intermediate steps.

Complexity Analysis:

Mention time complexity and space complexity with justification.

Advantages, Disadvantages and Applications:

Discuss practical uses and limitations.

Conclusion:

Summarize the importance of the algorithm and where it is used.